

Degree: B. Tech. Semester: 3rd Sem — Course work
MID-SEMESTER EXAMINATION (Summer), June, 2026

Course Title: Probability Theory and Random Process
 Course Code: EAEPC303

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assume & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	A box B1 contains 10 white and 5 red balls and a box B2 contains 20 white and 20 red balls. A ball is drawn from each box. What is the probability that the ball from B1 will be white and the ball from B2 red?	2	CO3
1b	Ten independent binary pulses per second arrive at a receiver. The error (that is, a zero received as a one or vice versa) probability is 0.001. What is the probability of at least one error/second?	2	CO1
2a	Describe random variable, the cumulative distribution function (CDF) and the probability density function (PDF). Discuss the relation between the CDF and the PDF.	2	CO1
2b	In a coin-tossing experiment, the probability of head is p and the probability of tail is q . Define random variable X such that $X(\text{head}) = 1$, $X(\text{tail}) = 0$. Plot the graph of CDF and PDF of X .	2	CO1
3a	The probability density function of a random variable X is given by, $f(x) = \begin{cases} 8x & 0 \leq x \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$ Find the mean and variance of X .	2	CO1
3b	Find the characteristic function of a normal random variable, $X = N(\mu; \sigma^2)$.	2	CO1
4a	The joint PMF, $P_{xy}(i, j)$ of the random variables, X and Y are given below. $ \begin{array}{c c c c c} & & \text{j} & & \\ & & 0 & 1 & 2 & 3 \\ \text{i} & 0 & \frac{4}{84} & \frac{18}{84} & \frac{12}{84} & \frac{1}{84} \\ & 1 & \frac{12}{84} & \frac{24}{84} & \frac{6}{84} & 0 \\ & 2 & \frac{4}{84} & \frac{3}{84} & 0 & 0 \end{array} $ Are X and Y independent? Justify your answer.	2	CO1
4b	Explain marginal density function and joint density function.	2	CO1
5a	Explain covariance and correlation coefficient for two joint random variables X and Y .	2	CO1
5b	The joint CDF of a bivariate random variable (X, Y) is given by $F_{XY}(x, y) = \begin{cases} (1 - e^{-\alpha x})(1 - e^{-\beta y}) & x \geq 0, y \geq 0, \alpha, \beta > 0 \\ 0 & \text{otherwise} \end{cases}$ (a) Find the marginal CDFs of X and Y . (b) Show that X and Y are independent.	2	CO1